Repurposing Beyond the Nearest Analog: Debiasing Drug-Indication Models Against Chemical-Similarity Shortcuts

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Abstract. A zero-training nearest-neighbor chemical-similarity search already predicts a drug's indication well when that drug has a close analog among known treatments; across nine therapeutic classes in our data it averages 0.865 AUROC with no training at all. Its accuracy collapses, however, on drugs with no close known analog, down to 0.21–0.46 AUROC on four of the nine classes, which is exactly where a repurposing model needs to work. We adapt DebiasedDTA, a guide/predictor reweighting scheme built for drug-target binding-affinity prediction, to drug-indication prediction, training a three-view model (chemical structure, L1000 gene expression, and a text encoding of each drug's target function) to specifically target the signal this similarity shortcut cannot explain. The reweighting gives a clear benefit, under the ChEMBL label this benchmark is built on, on the class that motivated this study: epilepsy (+0.116 cold-bucket AUROC over an otherwise identical undebiased model), a small benefit on one other class, a modest cost on two more, and no usable signal either way on the remaining five, whose cold buckets turn out to contain too few positives per fold to support any claim. An external repoDB check of this number is underpowered (only 5 compounds overlap) and does not confirm it, so we report the delta as real under its original construction but not independently confirmed. A biological-interpretation check on a field the model never trains on, the Repurposing Hub's mechanism-of-action label, supports the broader pattern instead: the debiased model's top cold-bucket epilepsy candidates are enriched 23-fold for plausible antiseizure mechanisms over the candidate-pool base rate, a further lift over the already-large enrichment of the undebiased model. Adapting two debiasing methods from the literature, DRDM and TAPB, to this task shows our model still holds the best point estimate, but also surfaces that one of them looks strong mainly because of joint multi-indication training, not its own correction, a confound worth watching for generally. We report the mixed cross-class picture directly rather than as a uniform win, alongside a leakage audit confirming why literature about the drug itself cannot be used as a feature: PubMed abstracts predict a drug's eventual indication at 0.96 AUROC even before approval, almost entirely through mentions of other same-class drugs.

1 Introduction

Drug repurposing models are usually scored on how well they rank known treatments above everything else, and a large part of that score is explained by chemistry alone. In our data, a nearest-neighbor search over ECFP4 fingerprints, with no training, scoring each candidate by its maximum Tanimoto similarity to a known positive, already averages 0.865 AUROC across nine therapeutic classes (epilepsy, depression, antipsychotics, antibacterials, antidiabetics, antineoplastics, lipid-lowering drugs, NSAIDs, and antihypertensives). This is not a trivial baseline to beat, because much of it is not wrong: drugs really do cluster by mechanism, and a new analog of a known treatment really is a reasonable candidate. The question this paper asks is what happens on the drugs that do not have a close known analog, since those are the ones a repurposing model is actually for.

On that subset the picture changes sharply. Stratifying each class's test compounds by similarity to the training positives and looking at the bottom tertile, the nearest-neighbor search's AUROC falls to 0.21 on NSAIDs, 0.43 on epilepsy, 0.44 on antibacterials, and 0.46 on antineoplastics, each close to or below chance. A model that only reproduces this shortcut fails exactly where repurposing matters most. We ask whether a model can be trained specifically against this known bias, using a guide/predictor reweighting scheme from DebiasedDTA [15], originally built to debias binding-affinity prediction against chemical similarity, retargeted here at a different bias (chemical analog similarity to known indications, not structure-activity relationships) and a different task (indication classification, not affinity regression). We add two further signal sources beyond chemistry: L1000 gene-expression profiles, and a text view encoding what each drug's known target does, from UniProt function descriptions via a biomedical language model, connecting drugs through target function even when no training example saw that exact target with a positive label.

A model that uses text about the drug itself, rather than about its target, would appear to solve this problem trivially and incorrectly: we show that PubMed abstracts mentioning a drug predict its eventual indication at 0.96 AUROC even when restricted to literature published before that drug's own approval date, and that masking the obvious disease-defining terms barely changes this, because the remaining signal is mentions of other, already-approved drugs in the same class. We exclude this feature deliberately and report the audit that justifies doing so.

2 Related Work

Debiasing drug-target and drug-repurposing models. DebiasedDTA [15] introduced a two-stage guide/predictor framework for drug-target affinity prediction: a low-capacity guide is cross-validated on the training set to produce per-sample importance coefficients, and the predictor trains with sample weights interpolating from uniform toward those coefficients. It was evaluated only on binding-affinity regression, and its own discussion flags extension to other tasks as open.

A companion release, pydebiaseddta [2], made the framework reusable with arbitrary guide/predictor pairs. TAPB [13] debiases drug-target interaction prediction against protein-target popularity through an interventional correction built around a SMILES-BERT encoder and frozen ESM-2 target embeddings with no analogue on a task with no protein-target axis. The closest work on drug *repositioning* is DRDM [14], which reweights drug-disease graph edges during GNN training against node-degree bias on association benchmarks far denser than our nine-indication set. Neither method's code or numbers transfer as-is, so Section 5.7 implements and tests each mechanism adapted to our data rather than only discussing the mismatch. We are not aware of prior work using a chemical-similarity nearest-neighbor model as a DebiasedDTA-style guide for indication prediction.

Chemical similarity as an evaluation confound. Several papers observe that train-test compound similarity inflates apparent performance in molecular prediction: scaffold splits still leave substantial similarity [8], and similarity-aware evaluation shows standard splits dominate reported performance [23]. Guney [7] made this argument for drug repurposing specifically, showing many reported successes reduce to near-duplicate recovery. A companion study from this group [11] audits this confound on the same LINCS library and proposes no new model; this paper takes that measurement as a starting point and targets a training-time correction instead. The activity-cliff literature [19] shows similarity can also actively mislead, not just fail to inform.

Gene-expression and protein-function-text views. LINCS L1000 [18] gene-expression signatures are widely used for repurposing, including models predicting L1000-style profiles for unmeasured compounds [16]. TxGNN [10] is a prominent recent repurposing baseline, but its zero-shot construct is unseen diseases, not unseen scaffolds, so we treat it as a reference point rather than an apples-to-apples ablation. Chemprop [21,9] is the standard open-source message-passing baseline for molecular property prediction. Encoding free-text functional descriptions of proteins, not just identifiers, improves zero-shot generalization to unseen protein classes [1,12]; we apply the same idea to drug targets.

Label leakage in biomedical prediction. Label leakage through indirect textual or coded proxies for the outcome is a documented failure mode in clinical machine learning [5,17]. We show a version of this specific to our setting and use it to justify excluding drug-literature text as a feature.

3 Data

We use nine ChEMBL-derived [6,22] therapeutic-class indication labels (epilepsy, antibacterials, antidiabetics, antineoplastics, antipsychotics, betablockers as an antihypertensive proxy, depression, lipid-lowering drugs, NSAIDs) over the 9,878-compound LINCS [18] library, with ECFP4 fingerprints and consensus L1000

landmark-gene expression for every compound. Target annotations come from the Drug Repurposing Hub [4], joined by name: 2,290 compounds carry a non-null target field, spanning 1,392 unique human target genes. For each gene we fetch its UniProt [20] function-description text from the UniProt REST API; 1,385 of 1,392 genes (99.5%) return a usable comment, giving a text embedding for 2,286 of the 2,290 target-annotated compounds. The remaining 7,588 compounds, with no Hub annotation, receive an explicit zero vector and an availability mask rather than a learned placeholder, so the model can tell "no target signal" from "target signal of zero."

4 Method

Each compound is scored by a guide and a predictor. The guide is the same nearest-neighbor Tanimoto score from Section 5.1: a one-feature logistic regression wrapping this score is cross-validated within each outer training fold, five repeats of five-fold inner cross-validation, to produce an out-of-fold importance coefficient per training example, normalized to mean one. Examples the guide already explains well get a low coefficient; examples it cannot explain get a high one. This is the DebiasedDTA [15] mechanism, retargeted from a binding-affinity guide/predictor pair to indication classification.

The predictor is a three-view gated fusion model. Each view, structure (ECFP4, 2,048-dimensional), expression (978-dimensional), and target-function text (PubMedBERT, 768-dimensional, with an availability mask), is encoded by its own small MLP, combined through a learned per-sample softmax gate restricted to the views available for that compound; the text view is excluded from the gate entirely for compounds with no annotation, rather than zeroed and left in. The model trains with a binary cross-entropy loss whose per-sample weights interpolate from uniform at epoch one to the guide's coefficients at the last, DebiasedDTA's convex-combination schedule. Each continuous view is standardized on training-fold statistics only; an earlier version fed unnormalized text and expression into the MLP and produced a text-only ablation below chance despite genuinely informative features, which fold-wise standardization fixed.

5 Experiments

5.1 Evaluation protocol

We use Murcko-scaffold-grouped stratified five-fold cross-validation, five seeds, for all nine classes, 225 outer folds per variant. Following DebiasedDTA's warm/cold framing, we stratify each fold's test compounds by Tanimoto similarity to that fold's training positives into warm (top tertile) and cold (bottom tertile) buckets, the more theoretically appropriate test: debiasing is designed to improve the cold, low-similarity cases, not an overall AUROC dominated by easy warm ones.

Table 1. Cross-class mean AUROC, pooled over 5 scaffold-CV seeds (per-class numbers in `results/headline_mean_std_auroc.csv`).

nn_tanimoto	lr_ecfp	xgb_ecfp	text_only	fusion_struct	fusion_nodebias	fusion_full
0.865	0.857	0.917	0.733	0.818	0.913	**0.913**

5.2 Main results

Table 1 reports cross-class mean AUROC for the zero-training nearest-neighbor guide, plain ECFP4 baselines, the target-text-only ablation, and the proposed three-view debiased fusion model, against an otherwise identical undebiased ablation.

The zero-training guide is already strong (0.865 mean AUROC), confirming near-analog similarity explains much of apparent accuracy; it is weakest on epilepsy and antineoplastics, the most scaffold-diverse classes, and strongest on antipsychotics and betablockers, where a few tight chemical series dominate. The proposed fusion model beats the guide on seven of nine classes, with the largest margins on antineoplastics (+0.171) and epilepsy (+0.119). At the pooled-AUROC level, debiasing itself (fusion_full against fusion_nodebias, an otherwise identical uniformly-weighted model) makes almost no difference, 0.913 against 0.913, expected, since pooled AUROC is dominated by the easy warm majority, not the metric debiasing targets.

5.3 Does debiasing help where it should: the cold bucket

Figure 1 shows AUROC separately for the warm and cold tertiles. The guide's own cold-bucket collapse is stark and class-dependent: 0.21 on NSAIDs, 0.44 on antibacterials, 0.43 on epilepsy, 0.46 on antineoplastics, each near or below chance, against 0.74–0.97 in the warm tertile of the same classes. This is the central premise of the paper, confirmed directly rather than assumed.

Whether fusion_full improves on fusion_nodebias specifically in the cold bucket is the real test, and pooling all five seeds before checking gives a misleading answer, since it hides how many cold-bucket positives back each number. Recomputing the delta per seed rather than pooled surfaces this directly (Table 2): three classes, antipsychotics, depression, betablockers, average under one cold-bucket positive per seed, making any AUROC on them uninterpretable regardless of value, and two more, NSAIDs and lipid, remain too data-starved for the sign of their delta to be distinguished from noise.

Of the four classes with enough cold-bucket positives per seed to say anything, epilepsy, the class that motivated this study, shows a clear and relatively reliable benefit from debiasing: a mean delta of +0.116 against a standard deviation of 0.058 across five seeds, roughly twice its own noise. Antidiabetic shows a small positive effect of similar reliability. Antibacterial and antineoplastic show modest negative effects, and antineoplastic's delta, despite having the largest cold-bucket sample of any class (26 positives per seed), is under one standard deviation from zero and so is not clearly distinguishable from noise either. The defensible claim

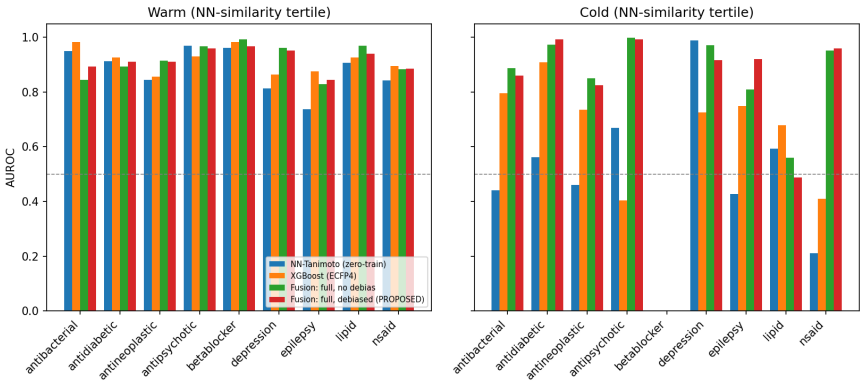


Fig. 1. AUROC by class, warm (top similarity tertile) and cold (bottom tertile) test compounds. Betablocker's cold tertile is omitted: its chemistry is dominated by one tight scaffold family, leaving too few genuinely low-similarity compounds to score.

Table 2. Cold-bucket debiasing delta (fusion_full − fusion_nodebias AUROC), computed per seed rather than pooled, with the mean number of cold-bucket positive examples available per seed. Classes with under ∼2 positives per seed on average cannot support a directional claim regardless of the delta shown.

Class	mean cold n_{pos}/seed	mean delta	std delta
epilepsy	6.2	**+0.116**	0.058
antidiabetic	2.2	+0.021	0.019
antineoplastic	26.0	−0.024	0.048
antibacterial	6.2	−0.028	0.022
nsaid	1.6 (2/5 seeds usable)	+0.034	0.050
lipid	2.2	−0.083	0.176
antipsychotic	0.4 (0/5 usable)	n/a	n/a
depression	0.2 (0/5 usable)	n/a	n/a
betablocker	0.0 (0/5 usable)	n/a	n/a

is narrower than either "debiasing helps across the board" or an unfiltered count of how many classes move in which direction: debiased reweighting gives a real, meaningful benefit on the motivating class and a smaller one on a second class, a modest and only weakly reliable cost on two more, and the remaining five classes simply do not have enough low-similarity examples in this dataset to test the method at all.

As a robustness check, we recomputed the same comparison using the bottom two quintiles (pooled across seeds, `results/similarity_quintile_auroc.csv`) instead of the bottom tertile, checked after writing the tertile numbers specifically to see whether they were a cutoff artifact. They are not: epilepsy's delta is +0.156 on 31 pooled cold-bucket positives, and every other class keeps the sign shown in Table 2.

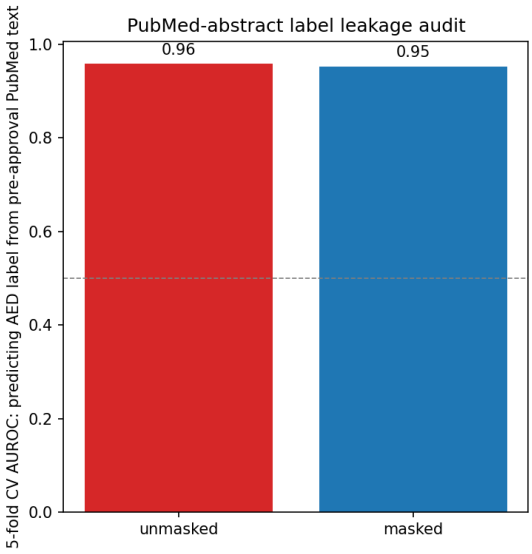


Fig. 2. Five-fold cross-validated AUROC for predicting a drug's eventual antiepileptic label from PubMed abstracts published strictly before its own approval date, with and without disease-defining terms masked ($n = 59$).

5.4 Leakage audit

A natural fourth view would be literature about the drug itself, not its target. We show why this is the wrong feature. For 30 known antiepileptic drugs and 29 non-antiepileptic drugs approved for antineoplastic or depression indications, we pulled PubMed abstracts mentioning each drug's name, restricted to strictly before its own approval year; 11 of 59 candidates, mostly pre-1960s drugs with sparse MEDLINE coverage, had no usable pre-approval text and were dropped. A TF-IDF logistic regression, five-fold cross-validated, predicts the eventual antiepileptic label from this pre-approval text at 0.959 AUROC (Figure 2). Masking obvious disease-defining terms (*epilep**, *seizur**, *anticonvulsant**, *antiepileptic**, *convulsant**, *convulsion**, *AEDs?*) barely changes this, 0.953, and the masked model's driving coefficients are almost entirely names of other antiepileptic drugs: lorazepam, midazolam, diazepam, carbamazepine, gabapentin, valproate, clonazepam, vigabatrin. The text predicts the label not through pharmacology but because papers about a future antiepileptic already co-mention other, already-approved antiepileptics as comparators, the same leakage mechanism documented for diagnosis codes in clinical prediction [5,17], transposed to drug-name co-occurrence, and why we exclude drug-literature text rather than rely on a held-out split to catch it.

5.5 Biological-interpretation positive control

A real AUROC gain could still come from a feature-set artifact rather than anything pharmacologically sensible, so we check whether the debiased model's top

Table 3. Top-50 cold-bucket MoA-keyword hit rate (mean ± std, 5 seeds), against the cold-bucket candidate base rate.

Model	Top-50 rate	
nn_tanimoto	0.004	±0.009
fusion_nodebias	0.228	±0.039
fusion_full (proposed)	**0.300**	±0.028
cold-bucket base rate	0.013	±0.001

cold-bucket picks make mechanistic sense, using a field the model never sees during training: the Broad Repurposing Hub's mechanism-of-action (MoA) string. Neither the text view, built from UniProt target-function text, nor any other input references this field. We first validate the check: of the 34 known antiepileptic drugs in our data, 18 resolve to a Hub MoA entry, and 14 of those (78%) contain one of *GABA*, *sodium channel*, *calcium channel*, *glutamate*, *potassium channel*, or *SV2A*, against a 7.8% base rate across the full MoA table, a roughly tenfold enrichment on drugs with a known mechanism.

Within the cold bucket and restricted to true negatives, excluding already-known antiepileptics so the check cannot rediscover training labels, we take each seed's top 50 candidates and compute this same keyword-hit rate.

The nearest-neighbor guide does worse than chance here, consistent with its cold-bucket collapse in Section 5: with no close structural neighbor, its top 50 carries no mechanistic signal. Both fusion variants recover a large, non-circular enrichment, 17 to 23 times the base rate, from structure, expression, and target-function text alone. Debiasing adds a further, consistent lift over the already-large undebiased enrichment (0.300 against 0.228, same direction across all five seeds), independent corroboration of the AUROC-based debiasing delta in Section 5 from a label source used nowhere in training or evaluation.

5.6 External validation of the epilepsy delta on an independent label source

The epilepsy delta rests entirely on ChEMBL-derived labels, the only indication source used anywhere in this paper. We checked it against repoDB [3], an approved-indication resource built from DrugCentral, independent of ChEMBL: 32 drugs carry an Approved epilepsy indication there, 23 of which resolve to a LINCS compound. Of those, 20 (87%) already carry our ChEMBL-derived positive label, a reassuring cross-source agreement; 3 are new positives ChEMBL missed, and 11 ChEMBL positives are unconfirmed by repoDB's older, differently-vocabularied snapshot.

Swapping the evaluation label on the already-frozen out-of-fold scores, no retraining, repoDB yields only 5 unique cold-bucket positives, the same 5 across all seeds, 4 already ChEMBL-positive: mostly the same drugs under a second label, not a fresh sample. The ChEMBL-label delta on this row set reproduces the headline $+0.115$; the repoDB-label delta is -0.002 ± 0.072. With 5 positives, one compound swapping (repoDB adds pyridoxine, drops newer AEDs it predates) moves the AUROC by the full effect size either way, a power limit, not a design

Table 4. Epilepsy cold-bucket AUROC (mean ± std, 5 seeds): adapted DRDM and TAPB baselines against this paper's own models.

Model	Cold-bucket AUROC
nn_tanimoto	0.425 ± 0.016
drdm_adapted	0.433 ± 0.190
tapb_adapted	0.852 ± 0.018
multitask_control (joint training, no debiasing)	0.863 ± 0.009
fusion_nodebias	0.799 ± 0.053
fusion_full (proposed)	**0.915** ± 0.052

choice. The broader premise survives: nn_tanimoto still collapses below chance (0.359), both fusion variants still reach ≈0.86 absolute AUROC. The +0.116 increment does not survive; we treat this as a limit on how much weight that number carries, not a refutation, since 5 positives cannot refute anything.

5.7 Empirical comparison against adapted DRDM and TAPB baselines

Neither debiasing method named in Related Work transfers to this task as its own architecture: DRDM's GNN was built for curated drug-disease benchmarks with thousands of diseases, and TAPB's correction is built around protein-target embeddings this task does not have. We instead implement the bias-correction *mechanism* of each, adapted to our nine-indication label set (full detail in `code/drdm_baseline.py` and `code/tapb_baseline.py`), and evaluate both on the identical epilepsy cold-bucket split used throughout. DRDM's annealed inverse-degree edge reweighting becomes a two-hop bipartite graph model over the nine label sets; TAPB's confounder-cluster backdoor adjustment becomes a clustering of the nine indications' positive-rate scores with per-cluster predictions combined by a shared classifier. Both share a feature-based drug encoder with the rest of the pipeline, trained jointly across all nine indications, since TAPB's correction requires that joint structure and DRDM gets the same setup rather than less data.

The DRDM-style baseline recovers no usable signal, 0.433 mean AUROC, indistinguishable from the nearest-neighbor guide and an order of magnitude noisier across seeds (std 0.190 against 0.016–0.058 elsewhere in this paper), consistent with a mechanism that needs a far denser association graph than nine indications and roughly 470 total edges can support. The TAPB-style baseline recovers strong, stable signal, 0.852, but a control isolating why, the same joint drug encoder with a plain linear head and no confounder-cluster correction, matches it almost exactly, 0.863. TAPB's correction adds nothing measurable beyond what joint multi-indication training already supplies; its apparent strength is a multi-task-learning effect, not evidence for its own mechanism. Our proposed model, with no joint multi-indication training, still reaches the highest point estimate, 0.915, though the margin over the multitask control sits within roughly one standard deviation of fusion_full's own seed-to-seed noise and should be read as suggestive, not decisive.

6 Discussion and Limitations

The central premise holds directly: a zero-training chemical-similarity search accounts for most of the apparent accuracy of a repurposing model, and its accuracy is close to or below chance on the novel-scaffold compounds a repurposing method is actually meant to find. Debiased reweighting against this shortcut is not, in our implementation, a uniform fix across therapeutic areas; it is a real, meaningfully-sized benefit on the class that motivated the study and a smaller one on a second class, with the remaining classes showing a modest, weakly-reliable cost or lacking enough low-similarity examples to test at all. We think this honest, uneven result, a debiasing method working clearly on its motivating case without working uniformly everywhere, is more useful than either extreme, so we report the per-class breakdown rather than one pooled number. The epilepsy delta holds under the ChEMBL label but is not confirmed by a repoDB check limited to 5 positive compounds (Section 5.6); the premise and the biological-interpretation enrichment both survive that check, while the +0.116 number itself remains real under its original construction and unresolved beyond it. Section 5.7's adapted-baseline comparison confirms the proposed model still holds the best point estimate against two debiasing methods from the literature, but also surfaces that one of them, TAPB, looks strong here mainly because of joint multi-indication training rather than its own correction, a confound worth watching for generally.

Several limitations should guide how much weight the current numbers carry. The fusion model encodes every view, including the 2,048-dimensional sparse ECFP4 view, through an MLP, while the strongest plain-chemistry baseline (xgb_ecfp) uses a tree ensemble, better suited to sparse binary features; Table 1 shows xgb_ecfp beating fusion_struct, the MLP-only ablation, on most classes. The fusion variants are controlled against each other, sharing one architecture, but not against xgb_ecfp, so part of the full model's margin over it could reflect the extra views rather than architecture alone. All results come from one library (LINCS) and one indication source (ChEMBL). Section 5.6's repoDB check of the epilepsy delta is underpowered (5 compounds), not confirmatory or disconfirming. Section 5.7's DRDM and TAPB comparisons adapt each mechanism to our label structure rather than reproducing its original architecture, since neither transfers as-is to a task with no protein-target axis (TAPB) or a graph this sparse (DRDM); Table 4 bounds what we could test, not either method's performance on its own benchmark. Chemprop ran on a reduced protocol (one seed, two folds, 15 epochs) since full D-MPNN training proved CPU-bound here, reflected in its fold variance (epilepsy 0.832 against 0.432). TxGNN was not run (no checkpoint in time; its zero-shot construct, unseen diseases, differs from ours regardless), and the guide's cross-validation used five repeats rather than DebiasedDTA's ten. Pooled bootstrap intervals treat non-independent resamples as independent, understating uncertainty; every claim rests on the per-seed mean and standard deviation instead. The leakage audit's 59-drug corpus is small by construction; its conclusion is clear, but its point estimate is not precise. Code: `https://github.com/edanmn/debiased-drug-repurposing`.

References

1. Akça, M.E., Uludoğan, G., Özgür, A., Baytaş, I.M.: STAR-GO: Improving protein function prediction by learning to hierarchically integrate ontology-informed semantic embeddings (2025)
2. Barsbey, M., Özçelik, R., Bağ, A., Atıl, B., Özgür, A., Özkırımlı, E.: A computational software for training robust drug-target affinity prediction models: pydebiaseddta. Journal of Computational Biology **30**(11), 1240–1245 (2023). https://doi.org/10.1089/cmb.2023.0194
3. Brown, A.S., Patel, C.J.: A standard database for drug repositioning. Scientific Data **4**, 170029 (2017). https://doi.org/10.1038/sdata.2017.29
4. Corsello, S.M., Bittker, J.A., Liu, Z., Gould, J., McCarren, P., Hirschman, J.E., Johnston, S.E., Vrcic, A., Wong, B., Khan, M., Asiedu, J., Narayan, R., Mader, C.C., Subramanian, A., Golub, T.R.: The drug repurposing hub: A next-generation drug library and information resource. Nature Medicine **23**(4), 405–408 (2017). https://doi.org/10.1038/nm.4306
5. Davis, S.E., Matheny, M.E., Balu, S., Sendak, M.P.: A framework for understanding label leakage in machine learning for health care. Journal of the American Medical Informatics Association **31**, 274–280 (2023). https://doi.org/10.1093/jamia/ocad178
6. Gaulton, A., Hersey, A., Nowotka, M., Bento, A.P., Chambers, J., Mendez, D., Mutowo, P., Atkinson, F., Bellis, L.J., Cibrián-Uhalte, E., Davies, M., Dedman, N., Karlsson, A., Magariños, M.P., Overington, J.P., Papadatos, G., Smit, I., Leach, A.R.: The ChEMBL database in 2017. Nucleic Acids Research **45**(D1), D945–D954 (2017). https://doi.org/10.1093/nar/gkw1074
7. Güney, E.: Reproducible drug repurposing: When similarity does not suffice. In: Pacific Symposium on Biocomputing (PSB). pp. 132–143 (2017)
8. Guo, Q., Hernandez-Hernandez, S., Ballester, P.J.: Scaffold splits overestimate virtual screening performance (2024)
9. Heid, E., Greenman, K.P., Chung, Y., Li, S.C., Graff, D.E., Vermeire, F.H., Wu, H., Green, W.H., McGill, C.J.: Chemprop: A machine learning package for chemical property prediction. Journal of Chemical Information and Modeling **64**(1), 9–17 (2024). https://doi.org/10.1021/acs.jcim.3c01250
10. Huang, K., Chandak, P., Wang, Q., Havaldar, S., Vaid, A., Leskovec, J., Nadkarni, G.N., Glicksberg, B.S., Gehlenborg, N., Zitnik, M.: A foundation model for clinician-centered drug repurposing. Nature Medicine (2024). https://doi.org/10.1038/s41591-024-03233-x
11. Kondadadi, R.K., McCreary, D.: How much do evaluation choices inflate transcriptomic drug repurposing? a measured audit on LINCS L1000 (2026), manuscript in preparation
12. Kulmanov, M., Hoehndorf, R.: DeepGOZero: Improving protein function prediction from sequence and zero-shot learning based on ontology axioms. Bioinformatics **38**(Suppl 1), i238–i245 (2022). https://doi.org/10.1093/bioinformatics/btac256
13. Lin, G., Zhang, X., Ren, Z., Zou, Q., Tiwari, P., Zhou, C., Ding, Y.: TAPB: An interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. Nature Communications **16** (2025). https://doi.org/10.1038/s41467-025-66915-1
14. Meng, Y., Wang, Y., Hu, X., Lu, C., Tang, X., Cui, F., Zeng, P., Yao, Y., Yang, J., Xu, J.: Adaptive debiasing learning for drug repositioning. Journal of Biomedical Informatics **167**, 104843 (2025). https://doi.org/10.1016/j.jbi.2025.104843

15. Özçelik, R., Bağ, A., Atıl, B., Barsbey, M., Özgür, A., Özkırımlı, E.: DebiasedDTA: A framework for improving the generalizability of drug-target affinity prediction models. Journal of Computational Biology **30**(11), 1226–1239 (2023). https://doi.org/10.1089/cmb.2023.0208
16. Pham, T.H., Qiu, Y., Zeng, J., Xie, L., Zhang, P.: A deep learning framework for high-throughput mechanism-driven phenotype compound screening and its application to COVID-19 drug repurposing. Nature Machine Intelligence **3**, 247–257 (2021). https://doi.org/10.1038/s42256-020-00285-9
17. Ramadan, B., Liu, M.C., Burkhart, M.C., Parker, W.F., Beaulieu-Jones, B.K.: Diagnostic codes in AI prediction models and label leakage of same-admission clinical outcomes. JAMA Network Open **8**(12), e2550454 (2025). https://doi.org/10.1001/jamanetworkopen.2025.50454
18. Subramanian, A., Narayan, R., Corsello, S.M., Peck, D.D., Natoli, T.E., Lu, X., Gould, J., Davis, J.F., Tubelli, A.A., Asiedu, J.K., Lahr, D.L., Hirschman, J.E., Liu, Z., Donahue, M., Julian, B., Khan, M., Wadden, D., Smith, I.C., Lam, D., Liberzon, A., Toder, C., Bagul, M., Orzechowski, M., Enache, O.M., Piccioni, F., Johnson, S.A., Lyons, N.J., Berger, A.H., Shamji, A.F., Brooks, A.N., Vrcic, A., Flynn, C., Rosains, J., Takeda, D.Y., Hu, R., Davison, D., Lamb, J., Ardlie, K., Hogstrom, L., Greenside, P., Gray, N.S., Clemons, P.A., Silver, S., Wu, X., Zhao, W.N., Read-Button, W., Wu, X., Haggarty, S.J., Ronco, L.V., Boehm, J.S., Schreiber, S.L., Doench, J.G., Bittker, J.A., Root, D.E., Wong, B., Golub, T.R.: A next generation connectivity map: L1000 platform and the first 1,000,000 profiles. Cell **171**(6), 1437–1452 (2017). https://doi.org/10.1016/j.cell.2017.10.049
19. van Tilborg, D., Alenicheva, A., Grisoni, F.: Exposing the limitations of molecular machine learning with activity cliffs. Journal of Chemical Information and Modeling **62**(23), 5938–5951 (2022). https://doi.org/10.1021/acs.jcim.2c01073
20. UniProt Consortium: UniProt: The universal protein knowledgebase in 2023. Nucleic Acids Research **51**(D1), D523–D531 (2023). https://doi.org/10.1093/nar/gkac1052
21. Yang, K., Swanson, K., Jin, W., Coley, C., Eiden, P., Gao, H., Guzman-Perez, A., Hopper, T., Kelley, B., Mathea, M., Palmer, A., Settels, V., Jaakkola, T., Jensen, K., Barzilay, R.: Analyzing learned molecular representations for property prediction. Journal of Chemical Information and Modeling **59**(8), 3370–3388 (2019). https://doi.org/10.1021/acs.jcim.9b00237
22. Zdrazil, B., Felix, E., Hunter, F., Manners, E.J., Blackshaw, J., Corbett, S., de Veij, M., Ioannidis, H., Lopez, D.M., Mosquera, J.F., Magarinos, M.P., Bosc, N., Arcila, R., Kizilören, T., Gaulton, A., Bento, A.P., Adasme, M.F., Monecke, P., Landrum, G.A., Leach, A.R.: The ChEMBL database in 2023: A drug discovery platform spanning multiple bioactivity data types and time periods. Nucleic Acids Research **52**(D1), D1180–D1192 (2024). https://doi.org/10.1093/nar/gkad1004
23. Zhang, C., Hu, Z., Jiang, C., Chen, W., Xu, J., Zhang, S.: Rethinking the generalization of drug-target affinity prediction algorithms via similarity-aware evaluation. In: International Conference on Learning Representations (ICLR) (2025)